DiffPoseNet++: Cheirality-Aware Contrastive Learning for Robust Pose Estimation

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Abstract

This study aims to implement, validate, and enhance DiffPoseNet, a deep learning-based visual odometry method that combines a pose estimator, a normal flow estimator, and a differentiable cheirality layer to reduce scene structure dependency and improve cross-dataset generalization. The research is conducted in three main phases: first, we implement NFlowNet and PoseNet components of the original DiffPoseNet architecture and validate them on the TartanAir dataset, where we aim to reproduce the reported performance metrics. Then, we focus on implementing self-supervised refinement using the cheirality layer to achieve and validate the original DiffPoseNet's pose prediction performance. Lastly, we enhance PoseNet's performance by incorporating architectural improvements alongside the self-supervised contrastive learning to improve feature discrimination and alignment across sequential frames, enhancing the model's ability to generalize in varied environments. These strategic enhancements aim to improve pose estimation accuracy and promote greater optimization stability during the self-supervised refinement phase. Comprehensive evaluations on the benchmark dataset are provided to quantify the performance improvements from our approaches. The code is available on GitHub at: https://github.com/ekaramustafa/diffposenet.git.

1. Introduction

Camera pose estimation is a fundamental component in a wide range of modern applications such as autonomous robotics, augmented reality, and simultaneous localization and mapping (SLAM) (Chen et al., 2023). Visual odometry, a key approach within this domain, estimates camera motion using only visual information from image sequences, without relying on external sensors or depth information. While traditional methods rely on optical flow or sparse feature matching to estimate camera motion, these approaches face challenges including noise sensitivity, scene structure dependence, and limited generalization capability (Sattler et al., 2018). *DiffPoseNet* (Parameshwara et al., 2022) addresses these limitations through an innovative architecture that integrates normal flow estimation and differentiable cheirality constraints to achieve robust and generalizable coarse-to-fine pose prediction. Normal flow, defined as the projection of optical flow onto the image gradient, serves as a guidance signal during self-supervised refinement, enabling the improvement of initial coarse pose estimates without the need for explicit depth computation. The cheirality layer enforces the depth positivity constraint to ensure physically plausible results.

However, despite its novel differentiable bi-level optimization approach, *DiffPoseNet* has demonstrated limited empirical performance by today's standards. Although it remains one of the few visual odometry studies that rely on the depth positivity constraint, its proposed architecture has not kept pace with recent advances in neural network design, such as recent self-supervised learning techniques and improvements in visual encoders that have significantly evolved over the past several years.

In this paper, we propose several significant architectural enhancements to DiffPoseNet, such as the integration of attention blocks and contrastive learning into the PoseNet component. We aim to demonstrate that well-designed modifications significantly improve the original performance of *DiffPoseNet*.

2. Related Works

The advent of deep learning revolutionized camera pose estimation, shifting the field from classical geometric methods to data-driven approaches. Early work by (Kendall et al., 2015) introduced PoseNet, the first end-to-end CNN to regress 6-DoF camera poses directly from RGB images. While efficient, PoseNet suffered from limited accuracy due

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to its reliance on monolithic regression. Subsequent efforts like Bayesian PoseNet (Kendall & Cipolla, 2016) addressed uncertainty modeling but retained fundamental limitations in scene geometry awareness.

Hybrid approaches emerged to combine learning with geometric principles. (Brachmann & Rother, 2017) proposed DSAC, which integrates differentiable RANSAC for robust PnP solving, while (Valada et al., 2018) introduced VLocNet, leveraging auxiliary tasks (e.g., odometry) to improve pose coherence. These works demonstrated that embedding geometric constraints—such as reprojection errors or temporal consistency—significantly boosts performance, as later echoed in (Shavit & Ferens, 2019)'s taxonomy of pose estimation methods. Our work builds on this insight by enforcing cheirality constraints to ensure physical plausibility.

(Camposeco et al., 2018) introduces a hybrid approach for camera pose estimation that combines 2D-3D (structure-based) and 2D-2D (structure-less) correspondences within a single RANSAC framework. Traditional methods rely exclusively on one type of correspondence, but the authors propose nine new minimal solvers that dynamically leverage both, improving robustness to noisy or incomplete 3D data.

(Melekhov et al., 2017) proposed a Siamese CNN for relative camera pose estimation, leveraging transfer learning and spatial pyramid pooling (SPP) to outperform traditional feature-based methods like SURF and ORB, particularly in challenging scenarios with textureless scenes or large viewpoint changes.

(En et al., 2018) propose RPNet, an end-to-end Siamese network for relative camera pose estimation, which directly infers full translation vectors (unlike traditional scale-ambiguous methods) and outperforms SIFT-based approaches on challenging textures. The system combines a PoseNet-inspired architecture with novel pose inference modules.

(Hayoz et al., 2023) is inspired by *DiffPoseNet*, the authors propose a camera pose estimation method that integrates the expressive power of deep neural networks with the robustness of geometric optimization. Similar to *DiffPoseNet*, their approach employs Deep Declarative Networks (DDNs) to enable gradient backpropagation through a pose optimization step that is not directly differentiable. By leveraging implicit differentiation, the method allows learning of the weighting parameters (θ_{2D}, θ_{3D}) in an end-to-end fashion, effectively combining deep learning with geometric reasoning.

Although the integration of contrastive learning (CL) to tackle tasks relevant to camera pose estimation has been well documented in recent years, most of these studies opted to introduce their own contrastive methodologies, depending on their goal, model architecture, and level of supervision.

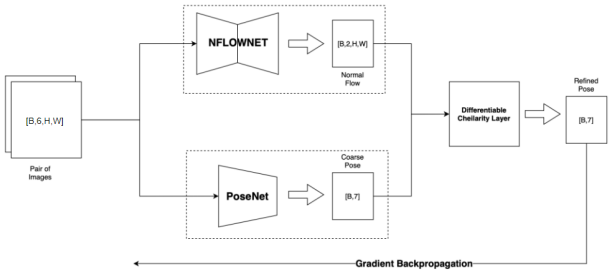


Figure 1. DiffPoseNet Architecture

For instance, (Zhang et al., 2022) utilizes pseudo labeling in its semi-supervised contrastive approach, operating on the encoded feature maps of input frames, for optical flow estimation. Likewise, despite derivating from *DiffPoseNet*'s optical flow representation for ego-motion estimation, (Lee et al., 2021) employs self-supervised contrastive learning via Contrastive Sample Consensus to improve upon the existing monocular video-based motion field estimation approaches. We intend to leverage the methodological diversity of contrastive learning-based research as a reference point for a potential CL implementation in our study.

Prior work in cross-view localization includes retrieval-based methods that employ Siamese architectures to match ground and aerial images. Recent approaches focus on fine-grained pose estimation, where contrastive learning aligns orientation-aware descriptors through techniques like Rolling & Matching. While existing methods either regress poses directly or use iterative optimization, they often struggle with ambiguous scenes or computational efficiency. As demonstrated in (Xia et al., 2023), advances this field by combining translational equivariance with multi-level contrastive matching to achieve robust joint localization and orientation estimation.

3. Approach

As demonstrated in Figure 1, *DiffPoseNet* consists of three main components: PoseNet, NFlowNet, and Cheirality Layer. PoseNet and NFlowNet are trained separately on the Tartan Air dataset. After the training of both models is completed, the weights of NFlowNet are frozen, and PoseNet is fine-tuned using the Cheirality Layer. This provides further refinement of PoseNet's predictions to improve its overall performance.

3.1. PoseNet

The PoseNet module produces an initial coarse estimation of the relative camera pose between the sequence of images. As demonstrated in Figure 2, it consists of a convolutional neural network (CNN) based on the pre-trained VGG-16

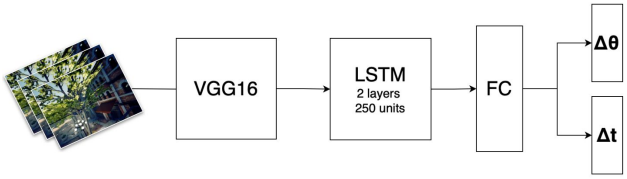


Figure 2. Original PoseNet Architecture

(Simonyan & Zisserman, 2015) to extract high-level spatial features. This is followed by two stacked long-short-term memory (LSTM) layers with 250 hidden units to capture temporal dependencies for the same sequence.

We formulate PoseNet mathematically for a better understanding. Let (I_t, I_{t+1}) denote a consecutive image pair. PoseNet extracts feature embeddings with a feature extractor $f_{\text{cnn}}(\cdot)$. This is followed by temporal extraction with an LSTM $f_{\text{lstm}}(\cdot)$, which creates an embedding:

$$z_c = f_{\text{lstm}}(f_{\text{cnn}}([I_t, I_{t+1}])) \in \mathbb{R}^d \tag{1}$$

where $[I_t, I_{t+1}]$ represents the input image pair.

The regression head $g(\cdot)$ then predicts the coarse relative pose from the embedding produced by LSTM:

$$(\hat{V}_c, \hat{\Omega}_c) = g(z_c) \tag{2}$$

where $\hat{V}_c \in \mathbb{R}^3$ and $\hat{\Omega}_c \in \mathbb{R}^3$ represent the estimated translation and rotation, respectively.

We introduce an improvement in PoseNet by replacing $\hat{\Omega}_c$ rotation with a quaternion representation to improve convergence and efficiency. Quaternions are more numerically stable and computationally efficient, and do not have discontinuities and singularities compared to other representations (Pavllo et al., 2018). Therefore, we reformulate the output of the regression head as follows.

$$(\hat{V}_c, \hat{q}_c) = g(z_c) \tag{3}$$

where $\hat{V}_c \in \mathbb{R}^3$ represents the estimated translation vector and $\hat{q}_c \in \mathbb{R}^4$ represents the estimated unit quaternion.

During training, we use the weighted L_2 loss between the predicted and ground-truth pose.

$$\mathcal{L}_{\text{pose}} = \lambda_t \|V^* - \hat{V}_c\|_2^2 + \lambda_q \left(1 - |\langle q^*, \hat{q}_c \rangle|\right) \tag{4}$$

where (V^*, q^*) are the ground-truth translation and unit quaternion rotation, $\langle \cdot, \cdot \rangle$ denotes the dot product, and λ_q and λ_t balances the contribution of rotation and translation loss.

3.2. PoseNet+

We improve PoseNet by introducing several key architectural modifications. In this section, we explain the reasons behind these changes and the issues we encountered, along with how we addressed them.

Although VGG-16 is efficient and lightweight, it is outdated and reduces the generalization ability of PoseNet. As demonstrated in Figure 3, to improve feature extraction, we replace VGG-16 with DINOv2 (Oquab et al., 2024). We also increase the complexity of the LSTM layer by adding more layers and making it bidirectional. This helps the model capture more complex temporal dependencies. The output features from the LSTM are then passed to a multi-head attention block with 8 heads. This allows the model to focus on the important parts of the LSTM output more flexibly. Furthermore, we design separate multilayer perceptrons with 3 hidden layers and dropout for the translation and quaternion/rotation heads. This helps the model learn each of them more effectively.

However, we encountered a critical issue which is the quaternion loss becomes very low in the early epochs although translation loss stays the same. This suggests that the model prioritizes learning quaternions. Even after increasing the number of epochs, the model could not reduce the translation loss on the validation set. To address this, we fine-tuned the loss balancing factors, as shown in the loss equation 4. However, this only delayed the convergence of quaternion learning and caused a slight increase in the total loss. Thinking that the authors might have used some crucial properties of rotation matrices to improve translation learning, we also tested them. As shown in Figure 4, both rotation and quaternion representations achieve different translation loss. The model performs better when quaternion representation is used. Furthermore, rotation matrices introduces significant noise during training. These findings support our primary motivation for quaternions. To improve translation learning, we applied uncertainty-based weighting (Kendall et al., 2018). This approach dynamically adjusts the loss weights using the intrinsic uncertainty of the model during training, allowing a better balance between λ_q and λ_t. The approach is formulated as follows:

$$\mathcal{L}_{\text{total}} = \frac{1}{2\sigma_t^2}\mathcal{L}_{\text{t}}(\theta) + \frac{1}{2\sigma_r^2}\mathcal{L}_{\text{q}}(\theta) + \frac{1}{2}\log\sigma_t^2 + \frac{1}{2}\log\sigma_q^2 \tag{5}$$

This approach did not lead to an improvement in translation loss. It also caused instability in the quaternion loss due to unconstrained variances in the default setup. We also limited the variances, but this did not improve the overall performance. As a result, we only introduced a learnable parameter for the translation loss without any variance terms. For the quaternion loss, we used a fixed balancing factor of 0.01. This approach is formulated as follows:

$$\mathcal{L}_{\text{total}} = \lambda_t^{(\text{learned})} \cdot \mathcal{L}_t + 0.01 \cdot \mathcal{L}_q \tag{6}$$

This setup gave us the best empirical results so far, although the improvement was minimal.

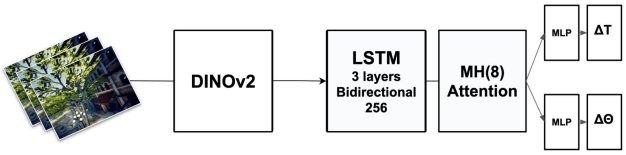


Figure 3. PoseNet+ Architecture

3.2.1. POSENET++

To address the learning issue in translation loss, we introduced a major change in the PoseNet architecture.

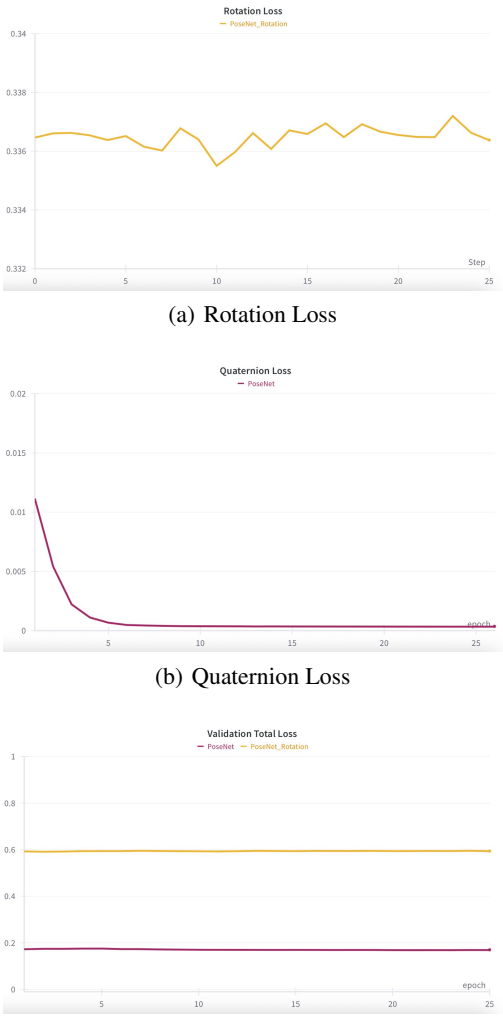
Both the original PoseNet and our version, PoseNet+, first extract spatial features using a pre-trained visual feature extractor and then use an LSTM to learn temporal features. Although this setup captures both spatial and temporal information, we hypothesize that it also requires proper alignment with contrastive learning to help the model distinguish between frames and their features. Without this alignment, the model may struggle to differentiate the frames, which produces poor performance in translation learning.

We explore two distinct approaches to incorporate contrastive learning into PoseNet+ architecture.

Approach 1: Contrastive Learning with Spatial and Temporal Features.
In this approach, we define positive pairs between the spatial features extracted by DINOv2 and the temporal features outputed after the LSTM and multi-head attention block. We apply contrastive loss to bring these feature representations closer together. The goal is to maintain alignment between spatial and temporal features, which allows the model to extract consistent and meaningful representations from the input sequence.

Approach 2: Contrastive Learning with Positive and Negative Samples.
This method defines contrastive pairs based on the structure of the TartanAir dataset, which consists of sequences of frames from various environments and videos. A positive pair is defined as a nearby frame sequence from the same video and environment. A negative pair is defined as a random frame sequence selected from a different environment and video. By applying contrastive loss in this setup, the model learns to distinguish between similar and dissimilar frame sequences, improving its temporal understanding and translation performance.

Regardless of the two approaches, integrating contrastive learning into the framework requires a modification to the loss function. Since the task is already multi-objective, we add one more term to the loss which is the contrastive loss.



(a) Rotation Loss

(b) Quaternion Loss

(c) Translation Loss

Figure 4. Rotation Matrices and Quaternion Experiments

We define the total loss as follows:

$$\mathcal{L}_{\text{total}} = \lambda_t^{(\text{learned})} \cdot \mathcal{L}_t + 0.01 \cdot \mathcal{L}_q + \lambda_c \cdot \mathcal{L}_c \tag{7}$$

Here, $\mathcal{L}_c$ represents the contrastive loss, and λ_c is a manually tuned hyperparameter that controls its contribution.

We experiment with different values of λ_c to observe how strength of contrastive learning affects the overall model performance.

3.3. NFlowNet

NFlowNet is a lightweight neural network designed to predict normal flow from consecutive image pairs, serving as a key supervised component in the DiffPoseNet pipeline for camera pose refinement. When the model is fully trained, its normal flow predictions are passed to the cheirality layer

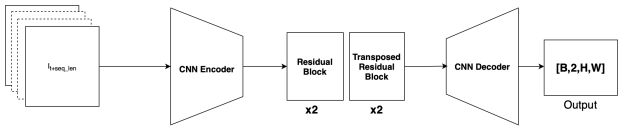


Figure 5. NFlowNet Architecture

for self-supervised pose refinement. By providing motion cues aligned with image structure, the predicted normal flow enables effective self-supervised refinement through the cheirality layer.

During both training and inference time, two consecutive video frames, concatenated along the channel dimension, are passed as the 6-channel inputs to the network. Subsequently, the U-Net outputs a 2-channel normal flow prediction, corresponding to the image pair. As demonstrated in 5, NFlowNet consists of a convolutional encoder-decoder architecture inspired by traditional U-Nets. However, the NFlowNet architecture does not include a bottleneck stage and skip connections between the encoder and decoder layers. Instead, residual connections exist only within individual encoder and decoder blocks. The architecture does not utilize any attention modules to maintain a lightweight design.

Let us mathematically formulate the NFlowNet to provide a clearer understanding of its internal structure and data flow. Let $[I_t, I_{t+1}] \in \mathbb{R}^{6\times H\times W}$ denote a consecutive frame pair, that is passed to the network. Following the cascade working principle of U-Nets, NFlowNet first extracts hierarchical spatial features with a CNN encoder module $f_{\text{enc}}(\cdot)$. These features are passed through stacked Residual Blocks $f_{\text{res}}(\cdot)$ and stacked Transposed Residual Blocks $f_{\text{tres}}(\cdot)$. Finally, it generates a dense normal flow map $\tilde{n} \in \mathbb{R}^{2\times H\times W}$ via the decoder module $f_{\text{dec}}(\cdot)$:

$$\tilde{n} = f_{\text{dec}}(\, f_{\text{tres}}(\, f_{\text{res}}(\, f_{\text{enc}}(\, [I_t, I_{t+1}]\,))))\tag{8}$$

During training, ℓ_2 loss between the predicted normal flow and the ground-truth normal flow n^* is used as the objective function:

$$\mathcal{L}_{\text{flow}} = \|n^* - \tilde{n}\|_2^2 \tag{9}$$

To evaluate NFlowNet's normal flow prediction quality, we employ Projection Endpoint Error (PEE), which measures how well our normal flow estimate $\tilde{n}$ aligns with the ground-truth optical flow u projected onto the image gradient ∇I:

$$PEE(u, \tilde{n}) = \|\tilde{n} - \frac{\nabla I}{\|\nabla I\|_2^2} u\| \tag{10}$$

3.4. Cheirality Layer

The final stage of DiffPoseNet is the **Cheirality Layer**. This is an optimization block to further refine the coarse pose estimation predicted by PoseNet. This layer enforces a geometric prior for camera pose estimation without explicit calculations on depth. The depth positivity constraint states that the scene must be in front of the camera to be visible. The Cheirality layer uses initial coarse pose $(\hat{V}_c, \hat{q}_c)$ and normal flow predictions $(\hat{n}_c)$ to minimize constraint errors. It uses implicit differentiation within a quasi-Newton solver (L-BFGS) to enable end-to-end differentiable training.

The refined pose (V_r, q_r) is used to fine-tune the weights of PoseNet during the second training stage, where NFlowNet's weights are frozen.

We formulate the Cheirality Layer mathematically as follows. Let n_x denote the normal flow magnitude at pixel x, and g_x be the unit vector along the image gradient direction. The flow in x is modeled as follows:

$$n_x = \frac{1}{Z_x}(g_x \cdot A)\mathbf{V} + (g_x \cdot B)\mathbf{q} \tag{11}$$

where $\mathbf{V}$ represents the translation vector, $\mathbf{q}$ represents the quaternion, Z_x is the depth of the 3D point, and A, B are matrices that relate the camera motion to the motion of the image plane.

We obtain the cheirality constraint:

$$\rho_x(V, q) = (g_x \cdot A\mathbf{V}) \cdot (n_x - (g_x \cdot B)\mathbf{q}) > 0 \tag{12}$$

which must hold for all visible points.

Following the original authors, we use GELU activation for modeling the cheirality constraint errors and minimize the following loss over all pixels:

$$\mathcal{L}_{\text{cheirality}} = \mathbb{E}\left[R(\rho_x(V, q))\right] \tag{13}$$

where $R(\cdot)$ represents the error function and the optimization is solved sequentially for V and q using the L-BFGS optimizer.

The cheirality layer can be expressed as a bi-level optimization problem as follows:

$$\begin{aligned}\underset{\tilde{P}_c}{\arg\min}\ \mathbb{E}\left(n_x - g_x \cdot \left(\left(\frac{n_x - (g_x \cdot B)\tilde{\Omega}_r}{(g_x \cdot A)\tilde{V}_r}\right) A\tilde{V}_c - B\tilde{\Omega}_c\right)\right) \\ \text{subject to}\ \underset{\tilde{P}_r}{\arg\min}\ \mathbb{E}\big(\mathcal{R}(\rho_x(\tilde{V}_c, \tilde{\Omega}_c))\big)\end{aligned} \tag{14}$$

4. Experimental Evaluation

4.1. Dataset

After implementing the components of *DiffPoseNet* (i.e., NFlowNet, PoseNet, and the Differentiable Cheirality Layer), we validate our results using the same dataset as

the original study (Parameshwara et al., 2022). Hence, we train both NFlowNet and PoseNet using the TartanAir dataset, which features a diverse set of indoor and outdoor photorealistic simulation environments with varying lighting, weather, and motion conditions (Wang et al., 2020a). Following the original implementation, we utilize videos with "Easy" trajectories in TartanAir's `(amusement, oldtown, neighborhood, office, office2, soulcity, japanesesealley, seasidetown)` environments for training. For evaluating NFlowNet, we utilize the `hospital` environment from TartanAir. Although the original paper also reports results on `abandonedfactory`, we were unable to include it in our evaluation due to download failures and incomplete dataset availability. PoseNet was evaluated on TartanAir's testing dataset prepared for the CVPR Visual SLAM challenge (Wang et al., 2020b). This setup ensures a fair and consistent comparison with the original work while highlighting the generalization capabilities of our implementation across varied and realistic environments.

4.2. Implementation Details

While this subset of environments constitutes a third of the full TartanAir dataset, we further reduced their effective size by disregarding depth and segmentation information, only using the images taken from the left camera with their corresponding positions, optical flows, and optical flow masks. Moreover, as the original paper failed to mention, samples with "Hard" trajectories are not included in the experimental dataset. Additionally, we opted to utilize the available optical flow masks when computing the ground truth normal flows for NFlowNet to avoid artifacts from occlusions or noisy regions; it is unclear whether the original paper employed the same masking strategy. By implementing our own dataset downloader, which better catered to our requirements, we were able to download and pre-process the 540 GB dataset.

4.2.1. Training and Testing Procedure

The NFlowNet training is carried out for 400 epochs with a batch size of 16. Adam optimizer with a fixed learning rate of 10^{-4} is used. Given the high computational cost of generating ground truth normal flows, we opt to precompute them prior to training and store them for efficient access during each epoch. Using five Nvidia Tesla V100 GPUs, the supervised NFlowNet training is completed in 33 hours. We train PoseNet in a supervised manner for 30 epochs, using four Nvidia Tesla V100 GPUs. We use Adam optimizer with a learning rate of 10^{-5}. To assess the performance gains from our architectural improvements, we conduct multiple training runs under identical conditions to ensure consistency and reliability in the results.



Figure 6. PoseNet+ vs PoseNet - Total Validation Loss

The quantitative assessment of our NFlowNet implementation is based on the Projection Endpoint Error (PEE), as used by the original paper and (Yuan et al., 2024). While we also incorporate the sequential flow loss from the implementation of (Luo et al., 2024), the associated performance metrics are not reported in this study, as the original DiffPoseNet paper did not include them, leaving no benchmark for direct comparison. The Absolute Trajectory Error is used to evaluate the original and improved PoseNet and *DiffPoseNet++*. Note that while the original implementation reports on PoseNet evaluation after performing self-supervised refinement, our PoseNet experimentation is limited by the first 30 epochs of coarse pose prediction training.

After fully training the NFlowNet and initializing PoseNet weights in 30 epochs to ensure adequate coarse pose prediction quality, we proceed with self-supervised refinement of PoseNet and PoseNet+ weights. Following the original implementation, we use the L-BFGS solver (Byrd et al., 2003) with strong Wolfe (Wolfe, 1969) line search function. The coarse pose predictions from PoseNet are used for optimizer initialization. The original study carried out 120 epochs for self-supervised refinement, resulting in significant PoseNet performance gains (Parameshwara et al., 2022).

4.3. Results

4.3.1. PoseNet Performance

As discussed in Section 3, we conduct a comprehensive set of experiments to observe the impact of the major architectural changes in PoseNet+ and PoseNet++ compared to the original PoseNet, in order to understand and verify the effectiveness of our proposals.

As demonstrated in Figure 6, our model significantly outperforms the original PoseNet. To analyze this improvement in more detail, we separately observe the translation and quaternion losses of PoseNet and PoseNet+.

PoseNet+ achieves lower translation loss compared to the

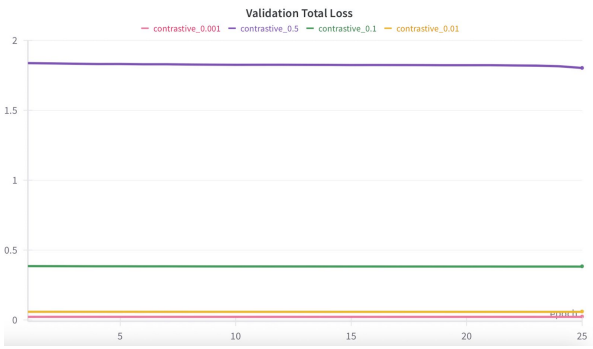


Figure 7. PoseNet++ Total Validation Loss

original implementation. The loss curve is almost identical to Figure 6 because the quaternion loss is very low and contributes minimally to the overall loss. However, our model performs slightly worse in quaternion loss during the initial epochs. However it eventually converges to an exact level. This pattern results from the learnable translation loss balancing factor, which we intentionally introduce to encourage the model to prioritize translation learning, at the cost of temporarily increasing quaternion loss. Moreover, the difference in quaternion loss is negligible. The maximum observed gap is only 0.00312.

To evaluate the effectiveness of contrastive learning in improving translation performance, we conduct experiments using different values of the contrastive learning loss factor λ_c. As shown in Figure 7, the total loss increases as λ_c becomes larger. However, the translation loss remains constant for different values of λ_c. The increase in total loss stems from the fact that the model regularizing quaternion learning more as the contrastive loss factor λ_c increases. Compared to our baseline PoseNet+ model, we conclude that contrastive learning does not produce any improvement in the overall performance.

Although we produced results for Absolute Trajectory Errors (ATEs) on the MH sequences, as done in the original paper, we do not include them here. However, we can provide these results upon request. The main reason for not presenting them is that the original paper reports performance after 120 refinement epochs with a cheirality layer. Since we were unable to implement the cheirality layer due to the issues explained in the corresponding section, we decided not to include these results.

4.3.2. NFlowNet Performance

One of the primary challenges in verifying the original NFlowNet implementation was the lack of information regarding the pre-processing steps applied to the ground-truth normal flows. The authors do not share whether optical flow masks are used. Moreover, we do not know whether any motion thresholding was applied to the optical flows in the dataset. When directly calculated from the optical flows and image gradients, the ground-truth normal flows become extremely noisy. One common solution, which we adopted, is to clamp normal flow with negligible magnitudes to zero, producing significantly sharper normal flow maps. The trade-off is a potential loss of fine motion details, particularly in low-texture or low-motion regions.

Given the limited clarity regarding pre-processing steps, we have trained NFlowNet under various training configurations to achieve quantitative results comparable to the original ones. In Table 1, we compare our experimental PEE values, when using and not using optical flow masking, with those provided in the original paper (Parameshwara et al., 2022). Given the strong dependence of the PEE metric on the magnitude of the normal flow vectors, we conclude that even minor differences in pre-processing can lead to significant shifts in experimental performance, making exact reproduction of the original results practically impossible without access to detailed pre-processing information.

Method	*abandonedfactory*				*hospital*				Num.
	000	001	002	003	000	001	002	003	param.(M) ↓
LiteFlownet	2.56	1.82	1.93	2.15	3.17	2.68	2.45	1.93	5.37
PWC-Net	1.23	0.95	1.64	1.48	2.35	2.92	2.28	1.47	8.75
UnFlow	1.35	1.15	1.75	1.56	2.21	1.76	1.83	1.07	126.90
SelFlow	**0.67**	0.73	**0.52**	0.64	1.91	0.51	0.73	**0.65**	5.11
NFlowNet (Original)	0.72	**0.54**	0.57	**0.63**	**0.82**	**0.44**	**0.57**	0.71	**2.72**
NFlowNet (Masked)	-	-	-	-	0.29	0.36	0.23	0.34	2.72
NFlowNet (Unmasked)	-	-	-	-	2.73	1.92	2.24	1.99	2.72

Table 1. PEE (Projected Endpoint Error) ↓ of different state-of-the-art methods compared to our NFlowNet implementation when using masked and unmasked GT normal flows. Lower values indicate better alignment with the ground truth optical flow.

As shown in Figure 8, our predicted normal flow maps exhibit strong visual similarity to their respective ground-truth counterparts. Our NFlowNet implementation achieves not only desirable spatial consistency, but also effective preservation of fine motion boundaries. Importantly, it generalizes well across both indoor and outdoor environments, maintaining high visual fidelity in diverse scene conditions.

4.3.3. Cheirality Layer Performance

We implemented the cheirality layer as described in the original DiffPoseNet paper. However, we encountered significant challenges during the refinement step, i.e., the bottom-level optimization. Despite following the reference implementation closely, the refinement process consistently failed to produce meaningful updates to the pose. Specifically, attempts to optimize translation and rotation parameters either jointly or separately did not yield improvements.

Several possible sources of this issue were considered. One potential reason is the poor quality of coarse pose predictions provided by PoseNet, which might have misled the

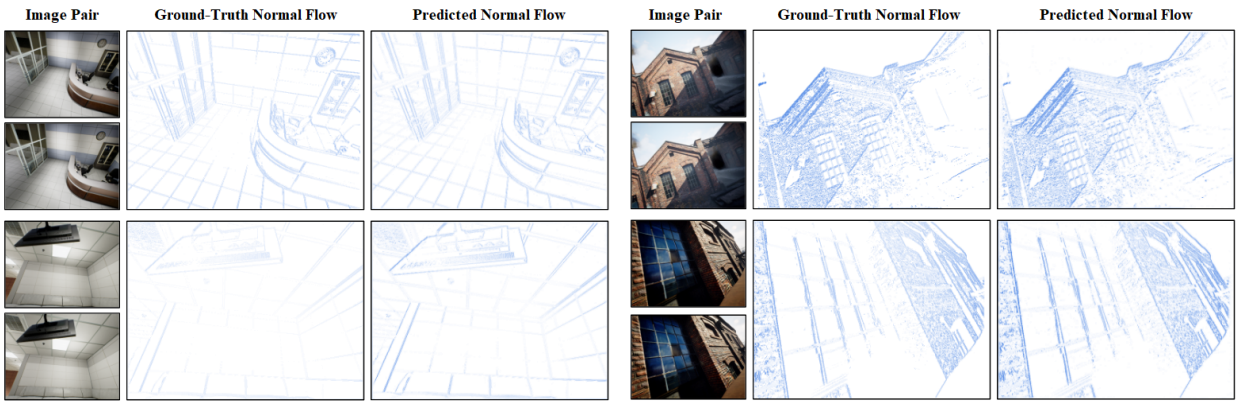


Figure 8. Qualitative comparison of predicted normal flow maps. The left block contains examples from indoor environments, while the right block shows samples from outdoor scenes. Our model captures fine structural details consistent with the ground truth.

optimizer. Another likely factor is the normal flow estimations produced by NFlowNet, which appeared to contain unusually high values in some regions. Additionally, errors in the calculation of the image gradient directions (g_x) could also have contributed to instability.

To isolate the problem, we replaced the coarse predictions with exact ground-truth values as input to the cheirality layer. However, even in this idealized setting, the refinement often failed. While in some image pairs certain pose elements were improved, other elements diverged significantly from the ground truth. We hypothesized that this behavior might be due to the optimizer getting stuck in local minima. Multiple runs with different initializations did not resolve the issue, and we were ultimately unable to achieve consistent refinement.

As a result of the failure in the bottom optimization, the upper-level optimization also failed to function, since it relies on a meaningful refinement signal from the lower level. However, when we simulated the bottom part as if it had produced exact ground truth poses, the upper-level optimization behaved correctly and updated PoseNet's parameters in a stable and meaningful way.

Due to these issues, the PoseNet model could not be refined through cheirality-based supervision. Consequently, we were unable to evaluate whether the translation inaccuracies could be corrected using cheirality constraints. Furthermore, since the original paper did not provide standalone outputs of the unrefined PoseNet, a direct comparison with our baseline model could not be performed.

5. Conclusion

In conclusion, we focused on three key components of DiffPoseNet: PoseNet, NFlowNet, and the Cheirality Layer. We successfully reproduced similar results and, in some cases, even surpassed the original performance in terms of different metrics and loss curves. For NFlowNet, we achieved competitive results compared to those reported in the original paper.

In PoseNet, we significantly improved performance by introducing major architectural changes, resulting in our enhanced version, PoseNet+. We also proposed a contrastive learning version of PoseNet+, which we call PoseNet++, to address the translation learning issue observed in PoseNet+. Although PoseNet++ did not outperform PoseNet+, it provided useful insights into how contrastive learning behaves in a multi-task learning setup. Due to unexpected issues with the Cheirality Layer, we were unable to refine the PoseNet architecture for 120 epochs as in the original work. In future work, we plan to implement the cheirality layer and resolve the current bugs, which, we believe, would allow us to produce Absolute Trajectory Error (ATE) results after cheilarity refinement. As we demonstrated in the experimental results, we believe that PoseNet+ has the promising potential to achieve significantly better results in both validation loss and ATE metrics. Moreover, NFlowNet lacks modern U-Net features such as skip connections and attention modules. In future works, we plan to incorporate a more expressive U-Net variant—with dense skip connections and self-attention to further improve its ability of capturing complex spatial dependencies.

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implementation of the project.

7. Contributions

Emircan Kocatürk conducted an extensive literature review on camera pose estimation, visual odometry, and related applications of contrastive learning. Implemented the cheirality layer.

Ebubekir Karamustafa developed TartanAir dataset downloader and loader. Introduced quaternion representation for better convergence. Implemented original PoseNet architecture. Proposed an improved version of PoseNet, PoseNet+ with major architectural modifications for both spatial and temporal feature extraction. Proposed two comprehensive contrastive learning strategies to address the translation learning issue observed in many experimental setups. Implemented PoseNet++ to integrate contrastive learning into PoseNet framework. Conducted 50+ quantitative and qualitative experiments using PoseNet, PoseNet+, PoseNet++.

İlkin Umut Melanlıoğlu assisted in the development of the TartanAir dataset downloader. Refined the downloaded dataset to make it fully functional and compatible with our pipeline. Using the Joblib Python library, pre-computed the GT normal flows to reduce the load on the CPU workers. Implemented the code and training for NFlowNet, using the accelerate library to enable fast and efficient multi-GPU training. Carried out training several times for masked and unmasked GT normal flows. Performed quantitative and qualitative assessment of the NFlowNet implementation. Assisted Ebubekir Karamustafa's integration into the HPC Cluster environment.

References

Brachmann, E. and Rother, C. Dsac-differentiable ransac for camera localization. In *CVPR*, 2017.

Byrd, R., Lu, P., Nocedal, J., and Zhu, C. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16, 02 2003. doi: 10.1137/0916069.

Camposeco, F., Cohen, A., Pollefeys, M., and Sattler, T. Hybrid camera pose estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 136–144, 2018.

Chen, C., Wang, B., Lu, C. X., Trigoni, N., and Markham, A. Deep learning for visual localization and mapping: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 2023.

En, S., Lechervy, A., and Jurie, F. Rpnet: An end-to-end network for relative camera pose estimation. In *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, pp. 0–0, 2018.

Hayoz, M., Hahne, C., Gallardo, M., Candinas, D., Kurmann, T., Allan, M., and Sznitman, R. Learning how to robustly estimate camera pose in endoscopic videos. *International journal of computer assisted radiology and surgery*, 18(7):1185–1192, 2023.

Kendall, A. and Cipolla, R. Modelling uncertainty in deep learning for camera relocalization. In *2016 IEEE international conference on Robotics and Automation (ICRA)*, pp. 4762–4769. IEEE, 2016.

Kendall, A., Grimes, M., and Cipolla, R. Posenet: A convolutional network for real-time 6-dof camera relocalization. *ICCV*, 2015.

Kendall, A., Gal, Y., and Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics, 2018. URL https://arxiv.org/abs/1705.07115.

Lee, S., Rameau, F., Pan, F., and Kweon, I. S. Attentive and contrastive learning for joint depth and motion field estimation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4862–4871, 2021.

Luo, A., Li, X., Yang, F., Liu, J., Fan, H., and Liu, S. Flowdiffuser: Advancing optical flow estimation with diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19167–19176, 2024.

Melekhov, I., Ylioinas, J., Kannala, J., and Rahtu, E. Relative camera pose estimation using convolutional neural networks. In *Advanced Concepts for Intelligent Vision Systems: 18th International Conference, ACIVS 2017, Antwerp, Belgium, September 18-21, 2017, Proceedings 18*, pp. 675–687. Springer, 2017.

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., and Bojanowski, P. Dinov2: Learning robust visual features without supervision, 2024. URL https://arxiv.org/abs/2304.07193.

Parameshwara, C. M., Hari, G., Fermüller, C., Sanket, N. J., and Aloimonos, Y. Diffposenet: Direct differentiable camera pose estimation. 2022. URL https://arxiv.org/abs/2203.11174.

Pavllo, D., Grangier, D., and Auli, M. Quaternet: A quaternion-based recurrent model for human motion. 2018. URL https://arxiv.org/abs/1805.06485.

Sattler, T., Maddern, W., Toft, C., Torii, A., Hammarstrand, L., Stenborg, E., Safari, D., Okutomi, M., Pollefeys, M., Sivic, J., et al. Benchmarking 6dof outdoor visual localization in changing conditions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8601–8610, 2018.

Shavit, Y. and Ferens, R. Introduction to camera pose estimation with deep learning. *arXiv preprint arXiv:1907.05272*, 2019. v3.

Simonyan, K. and Zisserman, A. Very deep convolutional networks for large-scale image recognition, 2015. URL https://arxiv.org/abs/1409.1556.

Valada, A., Radwan, N., and Burgard, W. Deep auxiliary learning for visual localization and odometry. *ICRA*, 2018.

Wang, W., Zhu, D., Wang, X., Hu, Y., Qiu, Y., Wang, C., Hu, Y., Kapoor, A., and Scherer, S. Tartanair: A dataset to push the limits of visual slam. 2020a. URL https://arxiv.org/abs/2003.14338.

Wang, W. et al. Tartanair: A dataset for simulating and evaluating visual slam in challenging conditions. https://sites.google.com/view/vislocslamcvpr2020/slam-challenge, 2020b. Accessed: 2025-06-17.

Wolfe, P. Convergence conditions for ascent methods. *SIAM Review*, 11(2):226–235, 1969. doi: 10.1137/1011036. URL https://doi.org/10.1137/1011036.

Xia, Z., Booij, O., and Kooij, J. F. Convolutional cross-view pose estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3813–3831, 2023.

Yuan, D., Burner, L., Wu, J., Liu, M., Chen, J., Aloimonos, Y., and Fermüller, C. Learning normal flow directly from event neighborhoods. *arXiv preprint arXiv:2412.11284*, 2024.

Zhang, Z., Bansal, N., Cai, C., Ji, P., Yan, Q., Xu, X., and Xu, Y. Clip-flow: Contrastive learning by semi-supervised iterative pseudo labeling for optical flow estimation. *arXiv preprint arXiv:2210.14383*, 2022.